Apache Tika Cheat Sheet: Detection, Extraction and Pipes

Tika 4.1.0 (3.3.2 for 3.x), 2026-10-13, Java 17+. AI-assisted: verify against official docs

1. Lines, Java and artifacts

Line	Latest	Java	Status (2026-10-13)
4.x	4.1.0	17+	Active; no 4.2 release yet
3.x	3.3.2	11+	Maintenance; ends Jan 2027 (wiki) or Mar 2027 (roadmap)
2.x / 1.x	2.9.4 / 1.28.5	8	End of life; affected by the 2025 XXE CVEs

Artifact (`org.apache.tika`)	For
`tika-bom` 4.1.0	One version for all Tika artifacts (Boot does not manage Tika)
`tika-core`	Detection, `Parser`/`Detector`, `Metadata`, `TikaInputStream`, facade
`tika-parsers-standard-package` (`<type>pom</type>`)	All standard parsers and charset detectors
`tika-parser-pdf-module`, `-microsoft-`, `-html-` ...	One family each, for a smaller class path
`tika-serialization`	`tika-config.json` and `TikaLoader`
`tika-pipes-fork-parser`	`PipesForkParser` (parse in a separate JVM)
`tika-langdetect-charsoup`	Language detection (recommended)
`tika-vlm`, `tika-inference`	VLM recognizers, embeddings (experimental)

Keep `tika-core` and every parser module at one version (mixing them caused CVE-2025-66516).

2. Which way to run Tika?

Option	Isolation	Use for
`Tika` facade / `AutoDetectParser`	None: OOM, hang, crash hit your JVM	Trusted files, tests, tools
`PipesForkParser`	Forked JVM, restarted after a crash	Java app receiving untrusted files
tika-app	Optional `--fork`, pipes mode	CLI, scripts, batch folders
tika-server (REST, :9998)	Parses in forked JVMs	Any language, Docker, scaling
tika-grpc (:9090 image)	Forked pipes workers	gRPC clients, streaming
Tika Pipes	Forks + limits, fetcher to emitter	Millions of files, indexing, RAG

Official advice: consider a service first. In-process parsing is not a sandbox.

3. Facade one-liners

```
Tika tika = new Tika();                 // one shared instance
tika.setMaxStringLength(-1);            // default 100,000 chars
String t1 = tika.detect(Path.of("catalogue.pdf"));
String t2 = tika.detect("notes.txt");     // by name only
String txt = tika.parseToString(Path.of("manuscript.docx"));
try (Reader r = tika.parse(path, metadata)) { /* stream */ }
```

`detect(InputStream)` consumes the stream in 4.x: wrap it in `TikaInputStream` and `rewind()`.

4. AutoDetectParser skeleton

```
static final AutoDetectParser PARSER = new AutoDetectParser();
Metadata md = new Metadata();                    // per document
BodyContentHandler h = new BodyContentHandler(-1); // no limit
ParseContext ctx = new ParseContext();           // per parse
try (TikaInputStream tis =
        TikaInputStream.get(Path.of("manuscript.docx"), md)) {
    PARSER.parse(tis, h, md, ctx);   // 4.x: TikaInputStream
}
String text = h.toString();
String type = md.get(HttpHeaders.CONTENT_TYPE);
```

Catch `ZeroByteFileException`, `EncryptedDocumentException` (supply a `PasswordProvider`), `WriteLimitReachedException` and `TikaException` per file. `tk:parsed-by` = `EmptyParser` means nothing parsed it.

5. Detection and media types

`DefaultDetector`: (1) `tk:content-type-override` wins; (2) `MimeTypes`: magic on the first 64 KiB, refined by the name glob (`tk:resource-name`) and the `Content-Type` hint; (3) container detectors (ZIP: OOXML, ODF, EPUB; OLE2: .doc, .xls, .msg); (4) `TextDetector` fallback; most specific wins.
`MediaTypeRegistry`: docx -> `application/x-tika-ooxml` -> `application/zip`; use `isInstanceOf`, `normalize` (aliases such as `application/x-pdf`). Custom types: `custom-mimetypes.xml` with `sub-class-of`. Validate uploads on detected bytes, never on the client's header; `application/octet-stream` = unknown.

6. Charset detection (4.x chain)

`bom-detector` -> `metadata-charset-detector` (HTTP charset) -> `html-encoding-detector` (`<meta charset>`) -> `mojibuster-encoding-detector` (UTF gates, Naive Bayes) -> `junk-filter-encoding-detector` arbitrates by decode quality (must stay last). Results: `tk:detected-encoding`, `tk:encoding-detector`, `tk:encoding-detection-trace`. The 3.x universal/ICU4J detectors need extra jars and explicit config. `AutoDetectReader` decodes for you.

7. Content handlers

`HANDLER_TYPE`	Handler	Names
`MARKDOWN` (4.x default)	`ToMarkdownContentHandler`	`markdown`, `md`
`TEXT`	`ToTextContentHandler`	`text`, `txt`
`BODY`	`BodyContentHandler` over text	`body`
`HTML` / `XML`	`ToHTMLContentHandler` / `ToXMLContentHandler`	`html` / `xml`
`IGNORE`	no-op (metadata only)	`ignore`

`new BodyContentHandler()` silently stops at 100,000 chars: pass a limit or `-1`. Build handlers with `BasicContentHandlerFactory(type, writeLimit)` or `newInstance(type, ctx)`; combine with `TeeContentHandler`. Parsers emit XHTML SAX events underneath.

8. Metadata keys (3.x to 4.x)

3.x	4.x
`resourceName`	`tk:resource-name`
`X-TIKA:Parsed-By`	`tk:parsed-by`
`X-TIKA:content`	`tk:content`
`X-TIKA:embedded_resource_path`	`tk:embedded-resource-path`
`X-TIKA:EXCEPTION:write_limit_reached`	`tk:exception:write-limit-reached`
`X-TIKA:digest:SHA256`	`tk:digest:SHA-256`
`pdf:PDFVersion`, `Message-From`	`pdf:pdf-version`, `message:from`

Unchanged: `Content-Type` (constant now `HttpHeaders.CONTENT_TYPE`), `dc:title`, `dc:creator`, `dcterms:created`, `xmpTPg:NPages`. Typed `Property` keys: `TikaCoreProperties`, `Office`, `PDF`, `XMPDM`.
`legacy-key-migration-filter` restores 3.x names during a migration; string writes to `tk:` keys throw.

9. Embedded documents

```
RecursiveParserWrapper w =
    new RecursiveParserWrapper(new AutoDetectParser());
var rh = new RecursiveParserWrapperHandler(
    new BasicContentHandlerFactory(HANDLER_TYPE.TEXT, -1));
w.parse(tis, rh, new Metadata(), new ParseContext());
List<Metadata> docs = rh.getMetadataList(); // container first
```

Same list: tika-server `/rmeta`, `tika-app -J`, Pipes `RMETA`. Each entry has `tk:content`, `tk:embedded-resource-path`, `tk:embedded-depth`. `embedded-limits`: `maxDepth`, `maxCount` (-1 = unlimited), `throwOnMaxDepth`/`throwOnMaxCount` (false: flag key instead of throwing).

10. Language detection

```
LanguageDetector d = LanguageDetector
    .getDefaultLanguageDetector().loadModels();
LanguageResult r = d.detect(text);  // resets first
r.getLanguage();  // ISO 639-3: "eng", "spa", "fra"
r.getConfidence(); r.isReasonablyCertain();
```

CharSoup (`tika-langdetect-charsoup`, recommended) or OpenNLP. Stateful: one per thread, `reset()` between documents. `charsoup-metadata-filter` writes `tk:detected-language`. 1.x `LanguageIdentifier` was removed in 2.0.

11. tika-config.json and TikaLoader

```
{ "parsers": [ {"pdf-parser": {"sortByPosition": true}},
               {"default-parser": {}} ],
  "parse-context": {
    "output-limits":   {"writeLimit": 1000000},
    "embedded-limits": {"maxDepth": 5, "maxCount": 1000},
    "timeout-limits":  {"totalTaskTimeoutMillis": 600000},
    "exception-reporting": {"level": "MESSAGE_REDACTED"} },
  "server": {"host": "127.0.0.1", "port": 9998} }
```

Top-level keys (strict; unknown key fails the load): `parsers`, `detectors`, `encoding-detectors`, `metadata-filters`, `renderers`, `text-recognizers`, `engines`, `inference`, `translator`, `content-handler-factory`, `auto-detect-parser`, `parse-context`, `presets`, `server`, `grpc`, `pipes`, `fetchers`, `emitters`, `pipes-iterator`, `pipes-reporters`, `plugin-roots`, `metadata-list`, `service-loader`, `xml-reader-utils`.

```
TikaLoader loader =
    TikaLoader.load(Path.of("tika-config.json")); // startup
Parser parser = loader.loadAutoDetectParser(); // shared
ParseContext ctx = loader.loadParseContext();  // per parse
```

Secrets only as `${env:NAME}` (4.1.0). Keep `default-parser` or only the named parsers load. 3.x XML: `tika-app --convert-config-xml-to-json=` (parsers only).

12. PDF and OCR

```
"engines": {"tesseract": {"tesseract-ocr-parser":
    {"language": "eng+spa", "timeoutMillis": 120000}}},
"text-recognizers": [ {"engine": "tesseract"} ],
"parse-context": {"pages": {"text": "AUTO",
    "render": {"dpi": 300}, "ocr": {"maxPages": 50}}}
```

`pages.text`: EXTRACT, AUTO (OCR pages with too little text), EXTRACT_AND_OCR, OCR, NONE. `"text-recognizers": []` = OCR off. `pdf-parser`: `sortByPosition`, `extractInlineImages`, `extractAnnotationText`, `maxPages`, `accessCheckMode`. Tesseract must be installed, or use the `apache/tika:4.1.0-1-full` image (Tesseract, GDAL, ImageMagick, ffmpeg).
VLM: `openai-vlm-parser`, `claude-vlm-parser`, `gemini-vlm-parser` as engines in `text-recognizers`; key via `${env:NAME}`; cost in `vlm:prompt-tokens`. **Inference**: `openai-embedding-engine` + `inference` bindings (TEXT, IMAGES, PAGES, MEDIA, task embed) write vectors to `tk:chunks`. Both experimental.

13. Extending: custom Parser / Detector

Implement `Parser` (not `AbstractParser`) or `Detector`; annotate `@TikaComponent(name = "bookshelf-order-parser")` with `tika-annotation-processor` (writes `META-INF/services` and `META-INF/tika/parsers.idx`), or hand-write `META-INF/services/org.apache.tika.parser.Parser` (no JSON name). A later parser for the same type wins. `ExternalParser` wraps a CLI tool.

14. Isolated parse: PipesForkParser

```
var cfg = new PipesForkParserConfig()   // one, shared
    .setTimeoutLimits(new TimeoutLimits(600_000, 5_000))
    .addJvmArg("-Xmx256m");
try (PipesForkParser fork = new PipesForkParser(cfg)) {
  try (TikaInputStream tis = TikaInputStream.get(path)) {
    PipesForkResult r = fork.parse(tis);
    r.getStatus();  // PARSE_SUCCESS, TIMEOUT, OOM, ...
  }
}
```

Crash, hang (`progressTimeoutMillis`) or OOM kills only the fork, which restarts.

15. tika-app flags (4.1.0)

Flag	Effect
`--md` (default), `-t`, `-x`, `-h`, `-T`, `-A`	Markdown, text, XHTML, HTML, main text, all text
`-m`, `-j`, `-J`, `-y`, `-r`	Metadata, JSON, recursive JSON (`/rmeta`), XMP, pretty
`-d`, `-l`, `--digest=sha256`	Detect type, language (ISO 639-3), digest
`-z` / `-Z`, `--extract-dir=`	Extract direct / all attachments (no `--unpack` flag)
`--config=FILE.json`	Load the JSON config with `TikaLoader`
`-f`, `--task-timeout`, `--progress-timeout`	Parse in a forked JVM, timeouts in ms
`-i dir -o dir`, `-n N`, `--handler=m`	Pipes mode: folders, forks, handler
`--list-parser-names`, `--list-detectors`	Component names for JSON

Removed: `--server`, `--client`, `-eX`, `-pX`, `--fork-timeout`, `--dump-*-config`.

16. tika-server

```
java -jar tika-server-standard-4.1.0.jar -c tika-config.json
docker run -d -p 127.0.0.1:9998:9998 apache/tika:4.1.0-1
curl -T file.pdf http://localhost:9998/rmeta/md
```

`/tika[/text|html|xml|md]`, `/rmeta[/{handler}]`, `/meta`, `/detect`, `/unpack[/all]`, `/language`, `/parsers`, `/version`, `/status` (opt-in), `/pipes`, `/async`. **429** = fork pool busy: honour `Retry-After` and retry. **503** = fork died (`TIMEOUT`, `OOM`, `UNSPECIFIED_CRASH`): retry once, then quarantine. A `-c` file needs a `"server"` block ({} is enough). `allowPipes` and `allowPerRequestConfig` default to false: keep them off.
tika-grpc: `apache/tika-grpc:4.1.0-1`, port 9090.

17. Tika Pipes

Roles: **iterator** (lists work, parent) -> **fetcher** (bytes, fork) -> **emitter** (writes result) + **reporter** (status). Unit: `FetchEmitTuple`. `parseMode`: `RMETA` (default), `CONCATENATE`, `CONTENT_ONLY`, `NO_PARSE`, `UNPACK`. `emitStrategy`: `DYNAMIC` (default), `EMIT_ALL`, `PASSBACK_ALL`. PF4J plugins: file system, S3, GCS, Azure Blob, HTTP, JDBC, CSV, JSON, Kafka, Solr, Elasticsearch, OpenSearch.

18. Limits and timeouts (parse-context)

Key	Default
`output-limits.writeLimit`	-1 (unlimited)
`output-limits.maxXmlDepth` / `maxPackageEntryDepth`	100 / 10
`output-limits.zipBombThreshold` / `zipBombRatio`	1,000,000 chars / 100
`embedded-limits.maxDepth` / `maxCount`	-1 / -1
`timeout-limits.totalTaskTimeoutMillis`	3,600,000 (1 h; 3.x: 5 min)
`timeout-limits.progressTimeoutMillis`	120,000 (stall, forks only)
`exception-reporting.level`	`FULL` (use `MESSAGE_REDACTED`)
`unpack-config.maxUnpackBytes`	10 GiB
`server.maxRequestSizeBytes`	1 GiB (413 above)

19. Security rules

- Parse untrusted files out of process (`PipesForkParser`, server, Pipes).
- tika-server and tika-grpc are not security boundaries: trusted network only, never public; bind `127.0.0.1`, auth at a gateway, NetworkPolicy.
- External commands (`ExternalParser`, `FileCommandDetector`): admins only, absolute paths.
- Secrets via `${env:NAME}`; upgrade past CVE-2025-54988 / CVE-2025-66516 (XFA XXE).

20. Spring Boot upload to tika-server

```
// RestClient tika = builder.baseUrl("http://tika:9998")
tika.put().uri("/rmeta/markdown")       // no Tika starter
  .contentType(MediaType.APPLICATION_OCTET_STREAM)
  .body(out -> Files.copy(file, out))   // stream from disk
  .exchange((req, res) ->
    switch (res.getStatusCode().value()) {
      case 200 -> RmetaMapper.toExtraction(res.bodyTo(LIST));
      case 429 -> retryAfter(res.getHeaders()); // retry
      case 503 -> throw new DocumentFailedException(...);
      default  -> throw new IllegalStateException(...); });
```

Spring AI `TikaDocumentReader` and LangChain4j `ApacheTikaDocumentParser` need Tika 3.x (`NoSuchMethodError` with 4.x).

21. Search and RAG

Solr: Solr Cell `/update/extract` with `tikaserver` backend, or `solr-emitter`. **Elasticsearch / OpenSearch**: `es-emitter` / `opensearch-emitter`, or the `attachment` processor. **Lucene**: index `tk:content` yourself. **RAG**: Markdown + `/rmeta`, one source per embedded document, `MarkdownChunker` or your splitter.

22. 1.x to 4.x

2.0: parser modules, Tika Pipes, typed metadata, no `LanguageIdentifier`. **3.0**: Java 11, jakarta, PDFBox 3, JSoup HTML. **4.0**: Java 17, zip distributions, `tika-config.json` + `TikaLoader` (no `TikaConfig`), `TikaInputStream` SPI, Markdown default, `tk:` keys, forked server, `PipesForkParser` (no `ForkParser`). **4.1**: `text-recognizers`, presets, metrics, `${env:}`.